8.2.1 Range

The difference between the largest value and the smallest value is called Range.

$$\text{Range } R = L - S$$

$$\text{Coefficient of range} = \frac{L-S}{L+S}$$

where L - Largest value; S - Smallest value

Progress Check

The range of first 10 prime numbers is _____

Example 8.1 Find the range and coefficient of range of the following data: 25, 67, 48, 53, 18, 39, 44.

Solution Largest value $L = 67$; Smallest value $S = 18$

$$\text{Range R} = L - S = 67 - 18 = 49$$

$$\text{Coefficient of range} = \frac{L-S}{L+S}$$

$$\text{Coefficient of range} = \frac{67-18}{67+18} = \frac{49}{85} = 0.576$$

Example 8.2 Find the range of the following distribution.

Age (in years)	16-18	18-20	20-22	22-24	24-26	26-28
Number of students	0	4	6	8	2	2

Solution Here Largest value $L = 28$

Smallest value $S = 16$

Range $R = L - S$

$R = 28 - 16 = 12$ Years

Note

If the frequency of initial class is zero, then the next class will be considered for the calculation of range.

Example 8.3 The range of a set of data is 13.67 and the largest value is 70.08. Find the smallest value.

Solution Range $R = 13.67$

Largest value $L = 70.08$

Range $R = L - S$

$13.67 = 70.08 - S$

$S = 70.08 - 13.67 = 56.41$

Therefore, the smallest value is 56.41.

Note

The range of a set of data does not give the clear idea about the dispersion of the data from measures of Central Tendency. For this, we need a measure which depend upon the deviation from the measures of Central Tendency.

8.2.2 Deviations from the mean

For a given data with n observations $x_1, x_2, \ldots x_n$, the deviations from the mean $\bar{x}$ are $x_1 - \bar{x},\ x_2 - \bar{x},\ \ldots,\ x_n - \bar{x}$.

8.2.3 Squares of deviations from the mean

The squares of deviations from the mean $\bar{x}$ of the observations $x_1, x_2, \ldots, x_n$ are $(x_1 - \bar{x})^2, (x_2 - \bar{x})^2, \ldots, (x_n - \bar{x})^2$ or $\sum_{i=1}^{n} (x_i - \bar{x})^2$

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